

Gafaruddin Shaik

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AI/ML Engineer | Generative AI Engineer | Data Scientist

[Linkedin](#)

PROFESSIONAL SUMMARY:

- AI/ML professional with **1.5+ years of industry experience** in data analytics, machine learning, and business intelligence
- Strong expertise in **Python, SQL, and data-driven problem solving** with hands-on experience in real-world projects
- Experience building **machine learning models for classification, prediction, and recommendation systems**
- Skilled in **data preprocessing, feature engineering, and model evaluation** using industry-standard techniques
- Hands-on experience with **Generative AI, LLMs, Retrieval-Augmented Generation (RAG), and prompt engineering**
- Developed AI applications including **chatbots, fake news detection systems, and recommendation engines**
- Experience working on **customer analytics and churn prediction systems** in a business environment
- Strong understanding of **NLP techniques including text preprocessing, tokenization, and feature extraction**
- Experience with **data visualization and dashboard creation** for business insights and decision-making
- Knowledge of **deep learning frameworks such as TensorFlow and PyTorch**
- Familiar with **big data technologies like Apache Spark, Kafka, and Airflow**
- Experience working with **relational and NoSQL databases** including PostgreSQL, MySQL, and MongoDB
- Strong ability to **translate business requirements into technical AI/ML solutions**
- Proven experience collaborating with **cross-functional teams and stakeholders**
- Exposure to **end-to-end ML lifecycle: data collection → modeling → evaluation → deployment concepts**
- Built and deployed **API-based and full-stack applications using JavaScript/TypeScript**
- Strong analytical and problem-solving skills with ability to work on complex datasets
- Passionate about building **scalable AI systems and intelligent applications**
- Continuously learning and applying **latest advancements in AI, ML, and Generative AI technologies**

TECHNICAL SKILLS:

- **Programming:** Python, SQL, Java, JavaScript, TypeScript
- **Machine Learning:** Scikit-learn, XGBoost, TensorFlow, PyTorch, NLP, Classification, Regression, Clustering
- **Generative AI:** LLMs, RAG, Prompt Engineering, Fine-Tuning, LangChain, Hugging Face Transformers
- **Data Engineering:** Apache Spark, Kafka, Airflow, Hadoop, ETL Pipelines, Data Warehousing
- **Databases:** PostgreSQL, MySQL, Oracle, MongoDB, Data Modeling
- **Visualization & BI:** Power BI, Tableau, Matplotlib, Seaborn, Plotly
- **Libraries & Tools:** Pandas, NumPy, Jupyter Notebook, Git, Docker, REST APIs
- **Cloud & MLOps:** AWS (S3, EC2), Model Deployment, CI/CD concepts, A/B Testing

PROFESSIONAL EXPERIENCE:

Role: Graduate Research Assistant (GRA)

Feb 2025 to till date

Company: Missouri University of Science and Technology

Project: Machine Learning-Based Data Analysis and Predictive Modeling

Roles & Responsibilities:

- Designed and deployed supervised ML models (XGBoost, Random Forest, Neural Networks) achieving 91%+ accuracy on structured research datasets with 500K+ records
- Built end-to-end ETL pipelines in Python and Apache Spark to preprocess, clean, and engineer features from large-scale datasets, reducing data prep time by 40%

- Applied hyperparameter tuning (GridSearchCV, Optuna) and cross-validation to optimize model performance; improved F1-score by 12% over baseline
- Implemented MLOps best practices including model versioning, experiment tracking with MLflow, and reproducible pipeline scripts
- Collaborated with faculty researchers to document findings and present AI-driven insights, contributing to 2 research reports
- Conducted statistical analysis and A/B testing frameworks to validate model predictions against real-world outcomes

Environment: Python, Apache Spark, Pandas, NumPy, Scikit-learn, XGBoost, TensorFlow/PyTorch, MLflow, Optuna, GridSearchCV, SQL, Jupyter Notebook, Git, Linux, Matplotlib/Seaborn

Role: Teaching Assistant (AI & Data Science)

Feb 2024 to Jan 2025

Company: Missouri University of Science and Technology

Project: Artificial Intelligence & Machine Learning Course Support

Roles & Responsibilities:

- Supported 30+ graduate students in building ML models using Python, Scikit-learn, and TensorFlow; held weekly office hours and code review sessions
- Guided students through core ML concepts including classification, regression, NLP, model evaluation, and cross-validation techniques
- Assisted in debugging and optimizing machine learning pipelines, reducing average student assignment completion time by 25%
- Evaluated 200+ assignments per semester, providing structured feedback on model design, feature engineering, and result interpretation

Environment: Python, Scikit-learn, TensorFlow, Pandas, NumPy, Jupyter Notebook, Matplotlib, Git

Role: Business Analyst

Jun 2022 to Nov 2023

Company: Tech Mahindra

Project: Customer Analytics & Business Intelligence Platform (Telecom/Banking Domain)

Roles & Responsibilities:

- Led customer analytics initiatives on a platform processing 5M+ customer record, applying SQL and Python to extract behavioral insights and identify churn risk segments
- Built Power BI dashboards tracking KPIs (churn rate, ARPU, engagement score) that improved reporting efficiency by 30% and reduced manual reporting from 8 hours to under 2 hours weekly
- Developed churn prediction model using Logistic Regression and XGBoost, achieving 85% recall; findings influenced retention campaign targeting 50K+ at-risk customers
- Designed and maintained ETL pipelines in Python to automate data ingestion, transformation, and loading into PostgreSQL data warehouse
- Performed feature engineering and statistical analysis to identify revenue trends; recommendations contributed to a 12% reduction in customer churn over two quarters
- Collaborated with cross-functional teams (product, marketing, engineering) to translate business requirements into data solutions and delivered stakeholder presentations

Environment: Python, SQL, Pandas, NumPy, Scikit-learn, XGBoost, Power BI, PostgreSQL, ETL Pipelines, Data Warehousing, Git, Jupyter Notebook

Role: Data Analyst Intern

Jan 2022 to Jun 2022

Company: Cognizant

Project: Data Processing & Reporting Automation System

Roles & Responsibilities:

- Developed Python and SQL scripts to automate data extraction and reporting workflows, reducing manual effort by 60% across 3 business reporting cycles
- Performed exploratory data analysis (EDA) on 1M+ row datasets; identified data quality issues that improved downstream model accuracy by 8%
- Wrote complex SQL queries for data retrieval, transformation, and aggregation; created 10+ reusable stored procedures adopted by the analytics team
- Designed and delivered weekly Excel and Power BI reports for 5 stakeholder teams, improving data visibility and decision turnaround time

Environment: Python, SQL, Pandas, NumPy, Excel, Power BI, PostgreSQL/MySQL, Data Analysis, EDA, Data Cleaning, Data Transformation, Git

Projects

RAG-Based Intelligent Chatbot

- Built a Retrieval-Augmented Generation (RAG) chatbot using LangChain, Hugging Face LLMs, and FAISS vector store; deployed via REST API with FastAPI backend serving 100+ queries/day

Fake News Detection System

- Developed an NLP-based fake news classifier using TF-IDF, BERT embeddings, and ensemble models; achieved 94% accuracy on a 40K-article benchmark dataset

Customer Churn Prediction Engine

- Built an end-to-end churn prediction pipeline with XGBoost and SHAP explainability; integrated into a Flask web app with REST API for real-time predictions

Recommendation System

- Designed a collaborative filtering and content-based recommendation engine using matrix factorization (SVD) and cosine similarity; evaluated with NDCG and MAP metrics

Education:

- Master's in information technology from MST, Rolla, MO, USA (GPA: 3.89/4.0) – 2025
- Bachelor of Technology (B. Tech) in Computer Science from GMRIT, Razam, AP, India (GPA: 3.0/4.0) – 2022